

Abstract

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Towards Effective Collaboration between Software Engineers and Data Scientists developing Machine Learning-Enabled Systems

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Integrating Machine Learning (ML) into existing systems is a challenge that has grown among several organizations. However, the development of ML-enabled systems encompasses several social and technical challenges, which must be addressed by actors from different fields of expertise working together. This paper has investigated how to enhance the collaboration between software engineers and data scientists in building these systems: software engineers and data scientists. We conducted two focus group sessions with data scientists and software engineers working on ML-enabled systems to assess the relevance of different recommendations for specific technical tasks. Our research indicates that collaboration between these actors is important for developing ML-enabled systems, especially when defining and ML model deployment. Participants provided insights on how recommendations depicted in the literature can improve collaboration during different tasks. For example, defining responsibilities for each team member and creating a shared vision can improve communication and overall performance. Our study contributes to a better understanding of effective collaboration between software engineers and data scientists creating ML-enabled systems.

Keywords: Machine Learning, ML-enabled System, Data Science, Software Engineering, Collaboration

1 INTRODUCTION

As clarified by Lewis et al. [2016], the integration of ML into existing systems may harm the ML-enabled systems if not done properly. However, suggested recommendations indicate that there is still a gap in assessing the relevance of tasks of ML-enabled systems.

Therefore, our study aims to investigate the perception of software engineers and data scientists on collaboration between SWEs and DSCs in building ML-enabled systems. For this purpose, we conducted two focus group sessions with data scientists and software engineers working on ML-enabled systems to assess the relevance of different recommendations for specific technical tasks. Our research indicates that collaboration between these actors is important for developing ML-enabled systems, especially when defining and ML model deployment. Participants provided insights on how recommendations depicted in the literature can improve collaboration during different tasks. For example, defining responsibilities for each team member and creating a shared vision can improve communication and overall performance. Our study contributes to a better understanding of effective collaboration between software engineers and data scientists creating ML-enabled systems.

To answer our question, we conducted two focus group sessions with experienced software engineers and data scientists. Following the guidelines of [2016], we promoted enriched discussions with DSCs on which tasks they consider relevant to collaborate. The results of the suggested recommendations' relevance allowed us to understand the perception of SWEs and DSCs on which tasks they consider relevant to collaborate. The results of the suggested recommendations' relevance allowed us to understand the perception of SWEs and DSCs on which tasks they consider relevant to collaborate.

Regarding the contribution, this paper contributes to the understanding of how to enhance the collaboration between SWEs and DSCs in building ML-enabled systems. This is essential for building ML-enabled systems. Following the guidelines of [2016], we promoted enriched discussions with DSCs on which tasks they consider relevant to collaborate. The results of the suggested recommendations' relevance allowed us to understand the perception of SWEs and DSCs on which tasks they consider relevant to collaborate.